



Higher Diploma in Computer Science

Computer Systems & Networks 2024



Contact Me

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Semester 2 2024

Computer Systems & Networks



logic · computer
organisation · os · networks
· interfaces · sensors

Database



entities · tables · rows · sql
· er · nosql

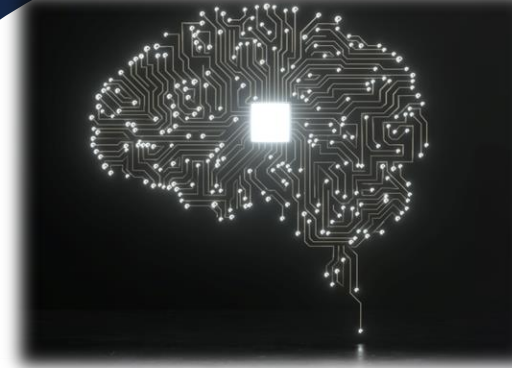
2024		S	M	T	W	T	F	S	Modules
Semester 2									
September									
	1	1	2	3	4	5	6	7	comp sys & database
	2	8	9	10	11	12	13	14	comp sys & database
	3	15	16	17	18	19	20	21	comp sys & database
	reading-week	22	23	24	25	26	27	28	
October									
	4	29	30	1	2	3	4	5	comp sys & database
	5	6	7	8	9	10	11	12	comp sys & database
	6	13	14	15	16	17	18	19	comp sys & database
	reading-week	20	21	22	23	24	25	26	
November									
	7	27	28	29	30	31	1	2	comp sys & database
	8	3	4	5	6	7	8	9	comp sys & database
	9	10	11	12	13	14	15	16	comp sys & database
	reading-week	17	18	19	20	21	22	23	
	10	24	25	26	27	28	29	30	comp sys & database
December									
	11	1	2	3	4	5	6	7	comp sys & database
	12	8	9	10	11	12	13	14	comp sys & database
		15	16	17	18	19	20	21	
		22	24	24	25	26	27	28	
January									
		29	30	1	2	3	4	5	

Weekly Webinar Schedule

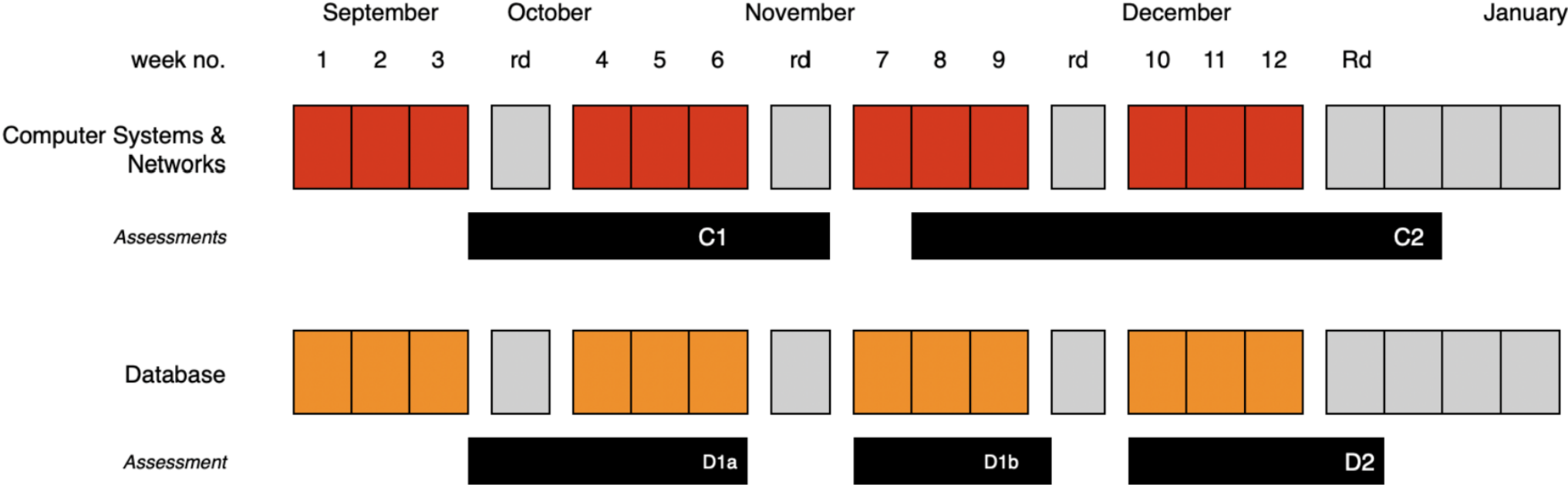
	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	
10:45						10:45
12:15	Computer Systems Webinar 12:15-2:00	Computer Systems Webinar 12:15-2:00 Database Webinar 2:15-2:00				12:15
2:00						13:45

Expectations

caroline.cahill@setu.ie



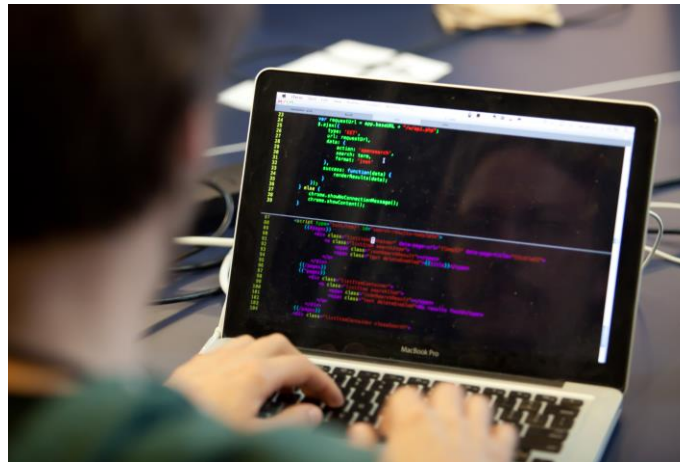
Semester 2 Assessment Schedule



TBCA for Caroline's Computer Systems



5% Cisco
NetAcad
chapter exams
(*due end Oct*)



25% Linux Shell scripting



5% InClass
Exercises

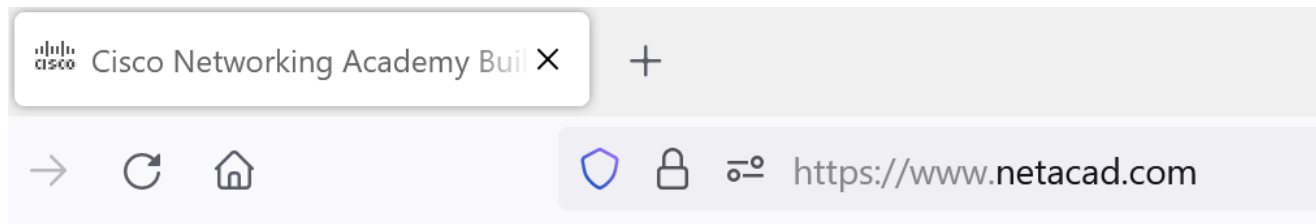
CA for Caroline's Computer Systems



5% Cisco
NetAcad
chapter exams
(*ongoing*)



Networking Academy

A screenshot of the Cisco Networking Academy login page. At the top, there is a "< Back" link and a language selector showing "US" and "EN". The Cisco logo is centered below the navigation bar. The main heading is "Log in". Below this, there are two input fields: "Email" with the value "caroline.cahill@setu.ie" and "Password" which is currently empty. The password field has a blue border and a small eye icon to toggle visibility.

Course Home

Gratis

Networking Academy

My NetAcad

Resources

Courses

Careers

More

Search

Notifications

Profile

Caroline

Home / I'm Teaching / HDip in CS 2024

Viewing as student. Switch back

HDip in CS 2024

Welcome to NDG Linux Essentials!

Front Page

Your progress

Find answers to many of your Networking Academy questions.

NetAcad Virtual Assistant

This entry level Linux course is offered by the Network Development Group (NDG). The skills taught in this course are applicable to a wide range of careers including networking, software development and Linux administration.

This course aligns to the Linux Professional Institute (LPI) Linux Essentials Professional Development Certificate. To learn more about this certificate, visit [LPI.org](#) (Links to an external site).

To Get Started:

1. Click on a course activity under a Module below.

2. Check the box to accept Terms and Conditions, click **Submit**.

3. Close the Terms and Conditions window (or tab), refresh the previous window (or tab), and reload the activity.

Need Help?

Click on the **Help & Resources** module below for Frequently Asked Questions. Please ensure you have read the course FAQ completely before contacting NDG Online Support.

NDG Linux Essentials version 2.21 (SLT)

NDG

www.netdevgroup.com

Help & Resources

Before You Get Started

Module 1 - Introduction to Linux

Before You Get Started

Chapter 1

Module 1 - Introduction to Linux

Chapter 1

Module 2 - Operating Systems

Chapter 2

Chapter 02 Exam

Module 3 - Working in Linux

Chapter 3

Chapter 03 Exam

Module 4 - Open Source Software and Licensing

Chapter 4

Chapter 04 Exam

Module 5 - Command Line Skills

Chapter 5

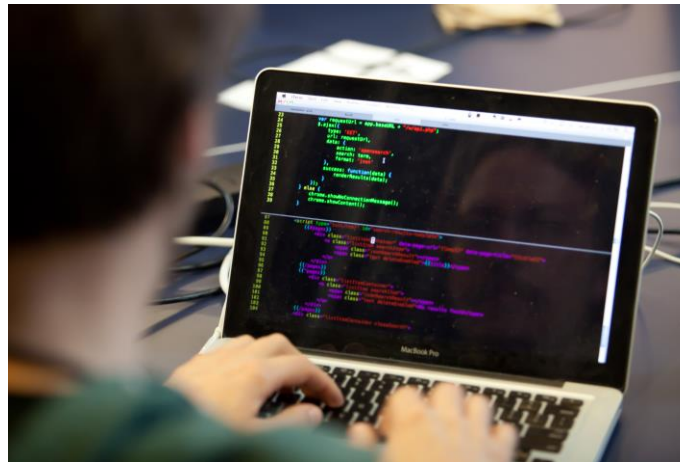
Lab 05

Chapter 05 Exam

CA for Caroline's Computer Systems



5% Cisco NetAcad
chapter exams (*due
end Oct*)



25% Linux Shell scripting assessment
(Given *end Sept*,
Due Sun 27th Oct '24)



5% InClass
Exercises

Assignments

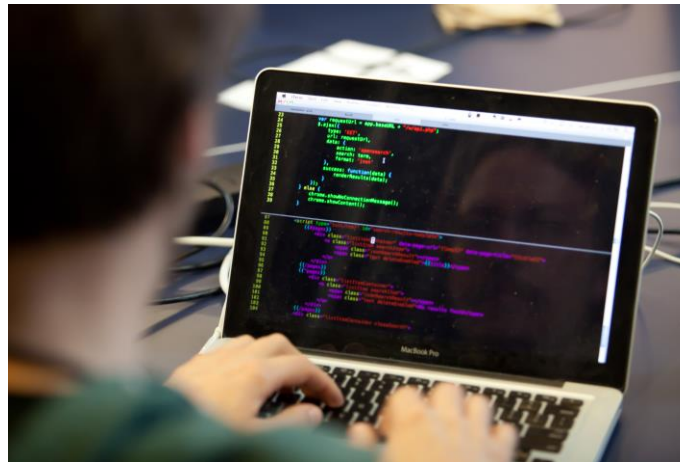
	September			October			November			December			January			
week no.	1	2	3	rd	4	5	6	rd	7	8	9	rd	10	11	12	Rd
Computer Systems & Networks																
Assessments	C1						C2									
Database																
Assessment	D1a						D1b						D2			

2023									
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CA for Caroline's Computer Systems



5% Cisco NetAcad
chapter exams (*due
end Oct*)



25% Linux Shell scripting

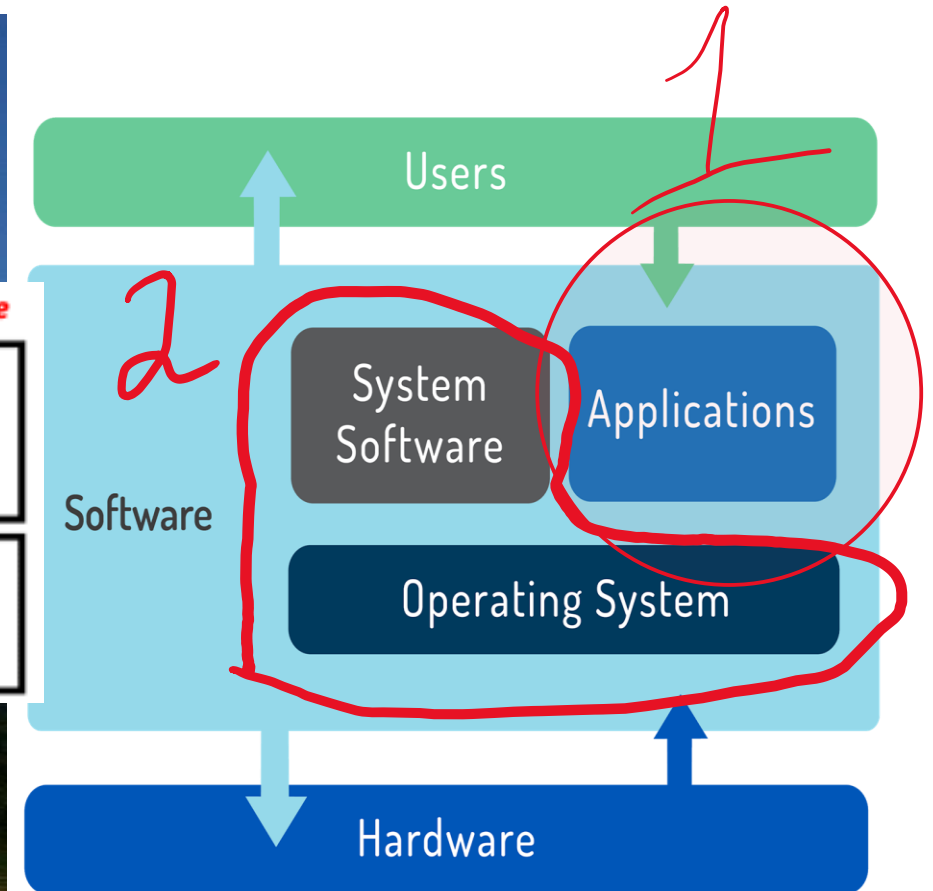
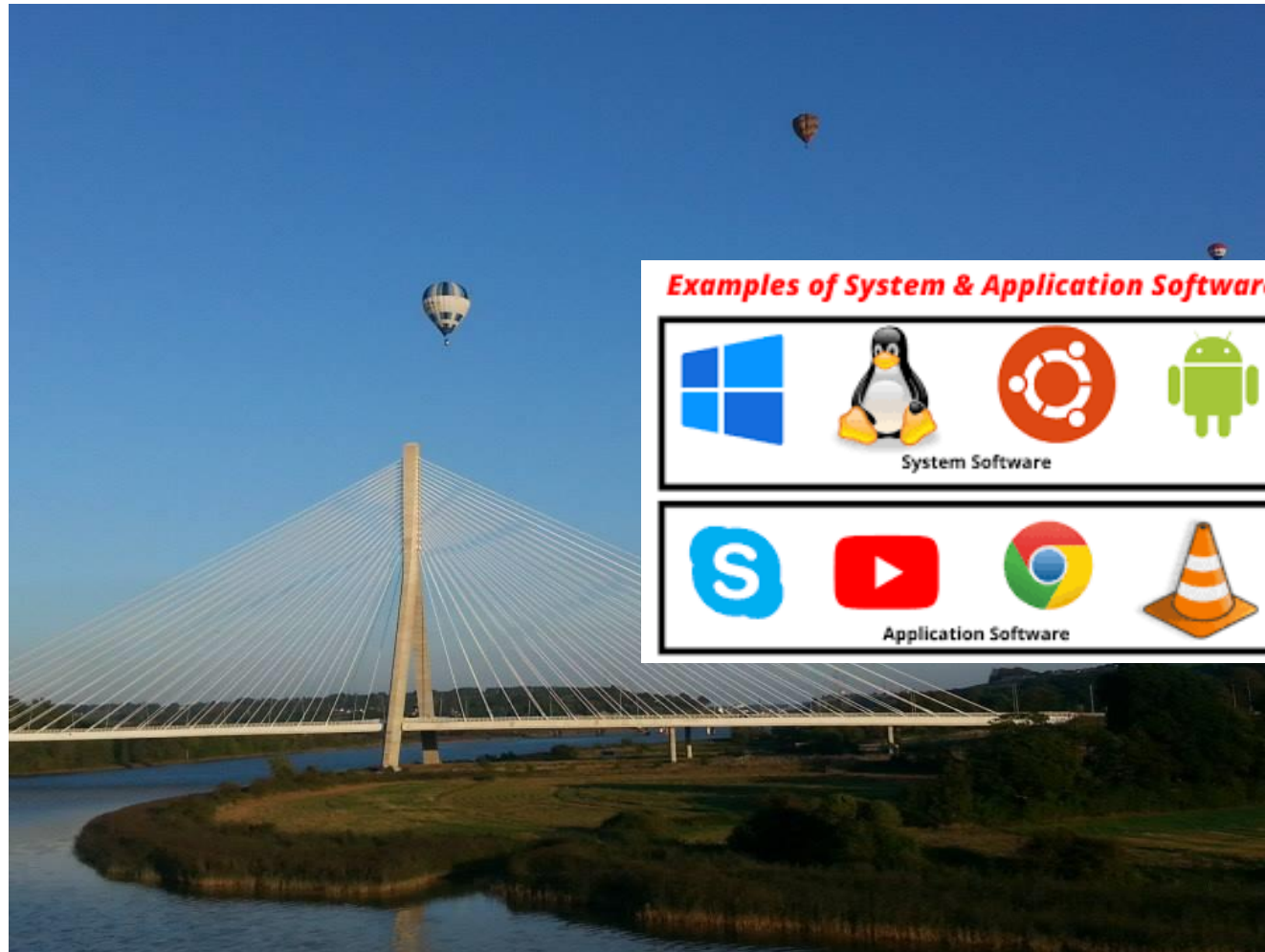


5% Classwork:
completed scripts
(during Sept)

Introduction to Operating Systems



Operating System (abbr. OS)



Decision Points when Considering which OS is best

- **Role:** Accessed by 1 user directly (desktop) or many users remotely (server)?
- **Function:** Does it need to run specific software?
- **Life Cycle:** What is the service lifetime?
- **Stability:** Are OS releases *beta* (not tested “in the wild”) or *stable* (tested)?
- **Compatibility:** Is it backwards compatible
- **Cost:** important factor for new systems

Introduction to Linux (as an OS)



Linux is Everywhere!

- Linux is everywhere, Linux jobs are everywhere, there is a shortage of Linux talent, and, Linux skills are in demand in just about every industry and job category.



Default (GPT-3.5)



why is it important to become familiar with shell scripting?



Becoming familiar with shell scripting offers several important benefits, especially for individuals working in IT, software development, data analysis, and system administration. Here are some reasons why it's important to learn shell scripting:

Copyright © 2019 Network Development

Linux is Open Source

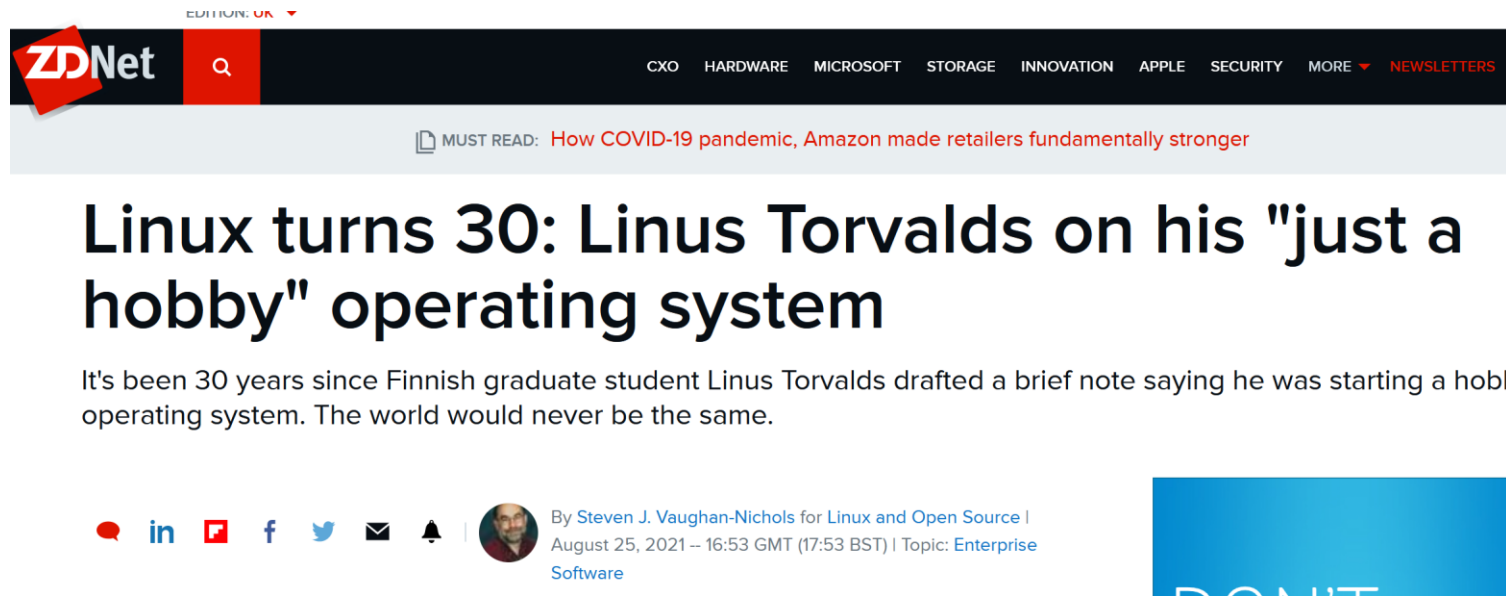
- The development of Linux closely parallels the rise of *open source software*.
- The open source philosophy is that

You have a right to obtain the software **source code** and

You have a right to **modify** it for your own use.

Linux is around a long time

- Linux started in 1991 as a hobby project by Linus Torvalds, a Finnish born computer scientist studying at the University of Helsinki.



Linux is a Kernel

- Linux means the *kernel* of the system, which is the central controller of everything that happens on the computer.
- Linux is a combination of software called **GNU/Linux**, which defines the *operating system*.
 - **GNU** is the *free software* that provides open source equivalents of many common UNIX commands.
 - The **Linux** part of this combination is the *Linux kernel*, which is the core of the operating system.

Linux Embraces the CLI

Two basic types of interfaces available that allow you to interact with the operating system



Linux Distros

A distribution refers to the Linux kernel, tools, and suite of applications that come bundled together.

Linux distros refer to an OS composed of open-source programs designed by numerous developers and programmers.



Lubuntu = Lightweight Ubuntu

- Ubuntu is one of the popular Linux distributions available for desktops and laptops
 - A new version of Ubuntu is released every April and October.
 - There's a thriving Ubuntu community online
-
- Refer back to our [May '24 onsite Prep Lab](#)

~~Linux~~ YOU Embrace the CLI

-
- Enter the Command

\$cal

- to view the built in calendar.

```
compsys@compsys-virtualbox:~$ cal
      August 2023
Su Mo Tu We Th Fr Sa
    1  2  3  4  5
 6  7  8  9 10 11 12
13 14 15 16 17 18 19
20 21 22 23 24 25 26
```

```
compsys@compsys-virtualbox:~$ cal
Command 'cal' not found, but can be installed with:
sudo apt install ncal
```

`$sudo apt install <<program>>`

```
compsys@compsys-virtualbox:~$ cal sept 2024
    September 2024
Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6  7
 8  9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30
```

```
$sudo apt update
```

Use this command to update the package database any time

~~Linux~~ YOU Embrace the CLI

- Try look up the current weather in Kilkenny using

`$ansiweather -l your_town_name`

```
compsys@compsys-virtualbox:~$ ansiweather -l kilkenny  
Weather in Kilkenny: 18 °C - UVI: 5.32 - Wind: 4.02 m/s WNW - Humidity: 75% - Pressure: 1016 hPa
```

Be familiar with the help **manual** available

What does the `-l` stand for in the command line instruction

```
$ansiweather -l kilkenny
```



~~Linux~~ YOU Embrace the CLI

- Uninstall ansiweather

\$sudo apt-get remove *program-name*

~~Linux~~ YOU Embrace the CLI

- Try out:

```
$cowsay Well hello there September!
```

Do you need to install it?

- Once finished, **uninstall**

Don't like cows??

```
$cowsay -l
```

```
$cowsay -f turkey Almost 100 days to Christmas, gobble!
```

These are examples of optional arguments that many CLI commands accept

~~Linux~~ YOU Embrace the CLI

- Try out:

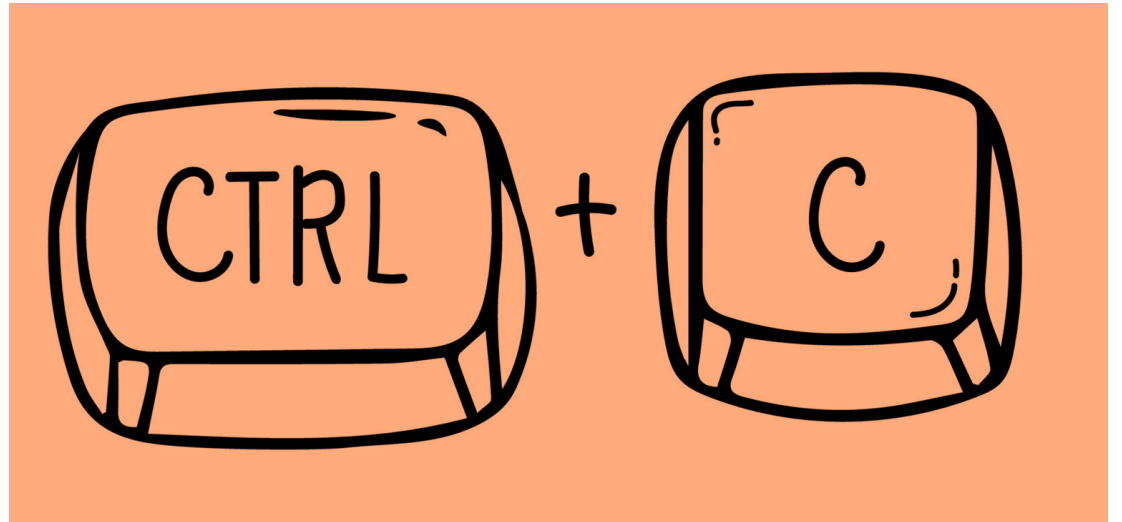
\$cmatrix

Now

\$figlet Lets go

And finally, what happens here
Why?

\$figlet Let's go



Try out this
weeks Lab

